

ECN 712 Fall 2025
EE Schlee
Problem Set II
Due Thursday, 4 September

1. Let the consumption set be $X = \mathbb{R}_+^L$ and $\succsim \subset X \times X$. Prove that (a) if $\succsim$ is *strictly* convex then the demand $d(p, w) = \{x \in B(p, w) \mid x \succsim y \text{ for every } y \in B(p, w)\}$ contains *at most* one element for each $(p, w) \gg 0$; and (b) if $\succsim$ is *convex* and *transitive*, then $d(p, w)$ is a convex set. Illustrate in a picture how each conclusion can fail if the associated convexity assumption is dropped.
2. Let a utility representation $u : \mathbb{R}_+^L \rightarrow \mathbb{R}$ be continuous, strictly increasing, strictly quasiconcave, and homogeneous of degree 1, so that preferences are homothetic. Prove that the corresponding demand d and the indirect utility v are both homogeneous of degree one in w for each $p \gg 0$. What is implied about the wealth expansion paths? What are the wealth elasticities of demand?
3. Consider the Cobb-Douglas utility function $u(x_1, x_2) = x_1^a x_2^b$ on $\mathbb{R}_+^2$ where $a, b > 0$.
 - (a) Is there any loss of generality in assuming that $a + b = 1$? If not, impose it in what follows. If so, then don't. In either case explain.
 - (b) Derive the demand function, ***justifying your steps carefully***. Let $s_\ell = p_\ell d_\ell / w$, good ℓ 's budget share. How does the budget share vary with prices and wealth?
 - (c) Are the preferences that u represents homothetic? Prove it. What are the wealth elasticities of demand, the own price elasticities of demand, and the cross price elasticities of demand?
 - (d) Calculate the indirect utility function. Verify that it is quasiconvex in p , nondecreasing in $(-p, w)$ and homogeneous of degree 0 in (p, w) .
4. Consider the *Stone-Geary* utility function defined on $X = \{x \in \mathbb{R}_+^L \mid x \geq x^0\}$ for some *non-zero* point x^0 in $\mathbb{R}_+^L$:

$$u(x) = \prod_{\ell=1}^L (x_\ell - x_\ell^0)^{\alpha_\ell},$$

where $\alpha_\ell > 0$ for $\ell = 1, \dots, L$; $\sum \alpha_\ell = 1$; and price-wealth pairs are restricted so that $w > \sum p_\ell x_\ell^0$.

- (a) Show *as concisely as possible* while *carefully justifying your steps* that the demand for good ℓ is given by (a change of variables and reflection on #3 would be most economical)

$$p_\ell d_\ell(p, w) = \alpha_\ell (w - p \cdot x^0) + p_\ell x_\ell^0.$$

(These demands are aptly called the *linear expenditure system*; their linearity made them popular in early empirical demand studies. Stone-Geary preferences have recently become popular again in macroeconomics to represent intertemporal consumption preferences.)
- (b) How does the budget share for good ℓ vary with prices and wealth? Are these preferences homothetic? How do you know?
5. Mas-Colell, Whinston and Green (from now on *MWG*), problem 3.D.4. Draw a picture of the wealth expansion path in part (c). And add two parts: (d) Solve for the demands when the function η from part (c) is given by $\eta(x_2) = 2\sqrt{x_2}$. *Carefully justify your steps*. (e) Are the preferences from part (d) homothetic? How do you know?

The following problems are **not** to be turned in:

6. In class we proved together that any indirect utility function (from a continuous, LNS utility) is quasiconvex. Prove that it is also (a) homogeneous of degree zero; and (b) nonincreasing in p and strictly increasing in w .
7. MWG, Exercise 3.C.6 (CES utility function; everyone should be familiar with this functional form, often still used in macro).